Tech Interview Assessment: X Visits = 1 Tree

Context

We want to build a system where **shop visits lead to planting trees for customers**.

A physical device (out of scope) detects when a user enters a shop and sends an event to a web service.

You will build that **web service**.

Requirements

Your service should:

- receive visit events from a device
- track visits associated to each customer
- store the last connection time for each customer
- after every configurable number of visits (**X visits = 1 tree**), increment a counter of **trees planted** for that customer
- expose a website with a frontend view showing the number of visits aggregated per hour

You are free to define:

- the API design
- the data model
- how data is stored and aggregated
- how the logic is structured

Keep the solution simple and pragmatic.

The persistence layer is up to you. It can be as simple as an in-memory variable or a database.

Deliverables

Provide a **GitHub repository** containing:

- the implementation of the service, including the backend and a simple frontend website
- a `README.md` explaining:
- how to run the project
- assumptions you made
- how to use the API

Notes

- You may use any language or framework
- Focus on clarity and simplicity over completeness
- You do not need to build the physical device
- The frontend can be simple, but it should be part of the solution

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Nice to Have (Optional)

- tests
- Docker setup
- API documentation
- architecture diagram
- short decision document explaining your technical choices
- any improvements you think are relevant

Goal

We want to understand how you:

- structure a small backend system
- design APIs and data models
- make technical decisions
- communicate your approach